

Roll No.

TBT-501

**B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023
RECOMBINANT DNA TECHNOLOGY**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write down the scope and applications of Recombinant DNA Technology. (CO1)
- (b) Give an overview of gene cloning with essential tool required in cloning. (CO1)
- (c) What is biosafety ? Write down different levels of biosafety with unique features. (CO1)
2. (a) Write brief notes on the following : (CO2)
 - (i) Restriction endonuclease
 - (ii) Blue white screening

P. T. O.

(2)

- (b) How a gene is cloned in plant cell for creating transgenic plant ? (CO2)
- (c) What is an artificial chromosome vector ? Discuss cloning and screening strategies in yeast artificial chromosome vector. (CO2)
- 3. (a) Write down different elements of an expression vector exemplifying any PET series vector. (CO3)
- (b) What is heterologous gene expression ? Discuss the role of codon optimization in it. (CO3)
- (c) Differentiate between gene library and *c*-DNA library including their applications. (CO3)
- 4. (a) Write down principle of Sanger's method of DNA sequencing. (CO4)
- (b) How can multiple copies of a DNA be made *in-vitro* ? What are different applications of PCR ? (CO4)
- (c) What is DNA probe ? How probes are labelled ? (CO4)
- 5. (a) Write down the methodology and applications of Gene therapy. (CO5)
- (b) What is CRISPR-CAS technology ? Discuss its application with reference to any congenital disease. (CO5)
- (c) Write down the process of production of recombinant Insulin. (CO5)

Roll No.

TBT-502

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

BIOPROCESS ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss *five* types of the products obtained in Bioprocess Industry.

(CO1)

(b) Explain Richard's rapid cooling method for del estimation. Compare this method with Graphical Method.

(CO1)

P. T. O.

- (c) Discuss time and temperature profile of sterilization. Compare Fibrous and Absolute air filters. (CO1)

2. (a) Aerobic degradation of an organic compound by xanthomonas species in waste water can be given by following reaction : (CO2)



(substrate)

(Biomass)

- (i) Determine a, b, c, d, and e if $Y_{x/s} = 0.4$ g cell/g substrate
- (ii) Determine degree of reduction of substrate and biomass.
- (b) Derive an expression for total and instantaneous Biomass as well as substrate concentration in Fed Batch Culture assuming quasi steady state conditions. (CO2)
- (c) Explain *five* factors of scale up and their significance along with formula applied in case of each factor. (CO2)
3. (a) Explain Dynamic gassing out and sulphite oxidation technique for estimation of KLa in bioreactors. (CO3)
- (b) Lactobacillus reuteri with $\mu_m = 0.5$ per hour is cultivated in a continuous culture under aerobic conditions where $D = 2.28 \text{ h}^{-1}$. The carbon and energy source in the feed is dextrose with initial concentration $S_0 = 2 \text{ g/l}$. The effluent dextrose concentration is desired to be $S = 0.1 \text{ g/l}$. If the growth rate is limited by oxygen transfer, by using the following information :

(3)

$Y_{x/s} = 0.45 \text{ g/g}$; $Y_{x/O_2} = 0.25 \text{ g/g}$ and $C^* = 8 \text{ mg/l}$. Determine the steady state biomass concentration (X) and specific rate of oxygen transfer. (CO3)

- (c) What are various components of a Feedback control loop ? Explain critical and non critical parameters of control in bioreactors. (CO3)
4. (a) Differentiate between Solid state and Submerged Fermentation. (CO4)
- (b) Describe fermentative production of Citric acid. Explain applications of Citric acid. (CO4)
- (c) Discuss fermentative production of Tetracycline. How do we recover it ? (CO4)
5. (a) What is Artificial Neural Network (ANN). Explain applications of ANN in bioprocess industry. (CO5)
- (b) What are different methods of enzyme immobilization ? What is the advantage of immobilization ? (CO5)
- (c) How do you screen industrially important microbes ? What is the application of lipases and amylases in industry ? (CO5)



Roll No.

TBT-503

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

BIOINFORMATICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Bioinformatics in your own words. Also elaborate genbank and fasta format. (CO1)
- (b) Discuss Windows and Linux operating system for running various bioinformatics applications. (CO1)
- (c) Elaborate any two real world examples where bioinformatics has been used successfully for solving biological problems. (CO1)
2. (a) Throw light on various dedicated search engines available for data/information retrieval. How do they make searching more efficient ? (CO2)

P. T. O.

- (b) Discuss any *two* techniques in biotechnology contributing data for bioinformatics. (CO2)
- (c) How has data management has been made easier with bioinformatics tools (software's and databases) ? (CO2)
3. (a) What do you understand by computational phylogenetic analyses ? What are the various components of phylogenetic tree ? Also write applications of phylogenetic analyses. (CO3)
- (b) Discuss how computational analyses can help in drug discovery and drug design. (CO3)
- (c) Elaborate various secondary and tertiary structures in RNA. Also throw light on various rules for structure prediction of RNA. (CO3)
4. (a) Draw a dot plot for the same sequence 'MAXISTAYAWAVATSIXAM', written horizontally and vertically. Also provide interpretation of the similarity observed between them. (CO4)
- (b) Discuss various networks that can be studied for understanding human diseases. (CO4)
- (c) What is genome editing ? What is the application of genome editing ? Also discuss computational genome editing. (CO4)

(3)

5. (a) Discuss intervention and also application of computational analyses in personalized medicines. (CO5)
- (b) What do you understand by genome wide association studies (GWAS)? Also discuss application of computational and mathematical analyses for GWAS studies. (CO5)
- (c) Discuss genome sequence analyses with computational tools and also applications of it. (CO5)

Roll No.

TBT-505-B

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

3D PRINTING AND DESIGN

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) How has 3D printing impacted rapid prototyping and product development cycles ? What are the limitations or challenges associated with 3D printing technology ? (CO1)
- (b) How has 3D printing contributed to the concept of distributed manufacturing and on-demand production ? (CO1)
- (c) What are some notable applications of 3D printing in fields such as healthcare, aerospace and automotive industries ? Support your answer with suitable examples. (CO1)

P. T. O.

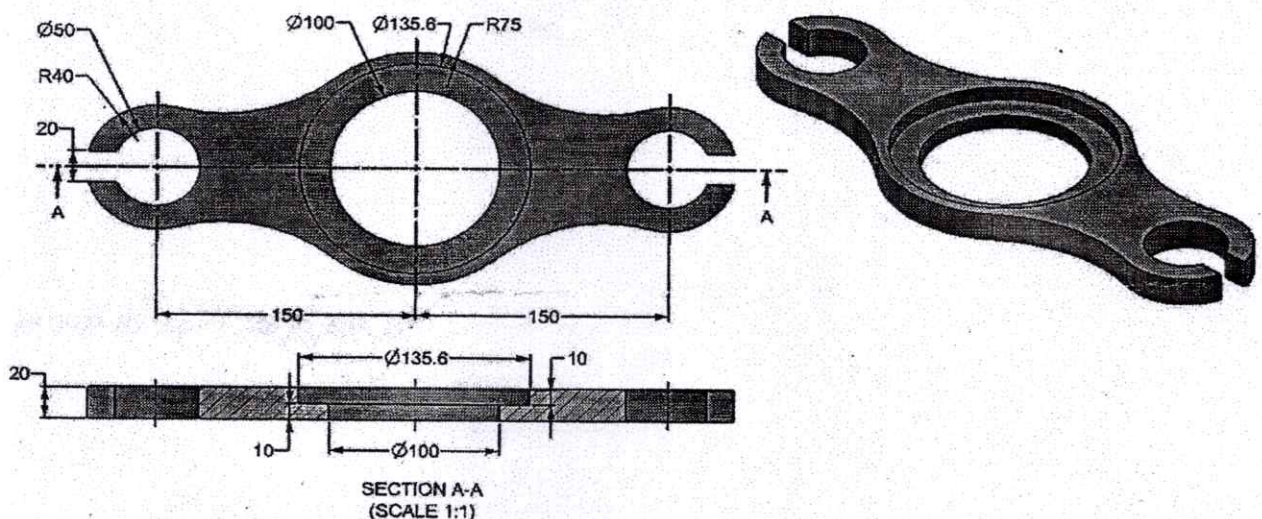
2. (a) How does the layer-by-layer deposition process occur in FDM printing and what is the significance of layer thickness in the final print quality ?
(CO2)
- (b) What types of materials are commonly used in FDM printing and how does the choice of material impact the final printed object ? (CO2)
- (c) How has FDM technology evolved over time and what are some recent advancements or innovations in this field ? (CO2)
3. (a) What does CAD stand for and how does Computer-Aided Design (CAD) contribute to the design and engineering processes ? Can you explain the fundamental principles of parametric design in CAD software and its significance in product development ? (CO3)
- (b) How does AutoCAD facilitate collaboration among design teams, especially in a multi-disciplinary engineering environment ? Can you discuss the importance of precision and accuracy in CAD design, and how AutoCAD ensures these qualities in the final output ? (CO3)
- (c) What are some common file formats used in CAD and how does interoperability between CAD systems impact data exchange in various industries ? In what ways has the evolution of CAD software, including AutoCAD, influenced the efficiency and innovation in product design and manufacturing processes ? (CO3)
4. (a) What are some advanced features or capabilities found in modern slicing software and how do they enhance the user experience and

printing outcomes ? In what ways does slicing software contribute to the efficiency of the 3D printing process, especially in terms of minimizing material waste and optimizing print time ? (CO4)

(b) What are the key advantages of SLA 3D printing over other additive manufacturing methods and in what industries is it particularly beneficial ? How does the layer thickness and resolution in SLA printing impact the final quality of printed objects ? (CO4)

(c) What are the primary applications of SLA 3D printing and how has it influenced the production of prototypes and end-use parts ? How does the SLS 3D printing process contribute to the creation of durable and functional parts, especially in industries like aerospace and automotive ? (CO4)

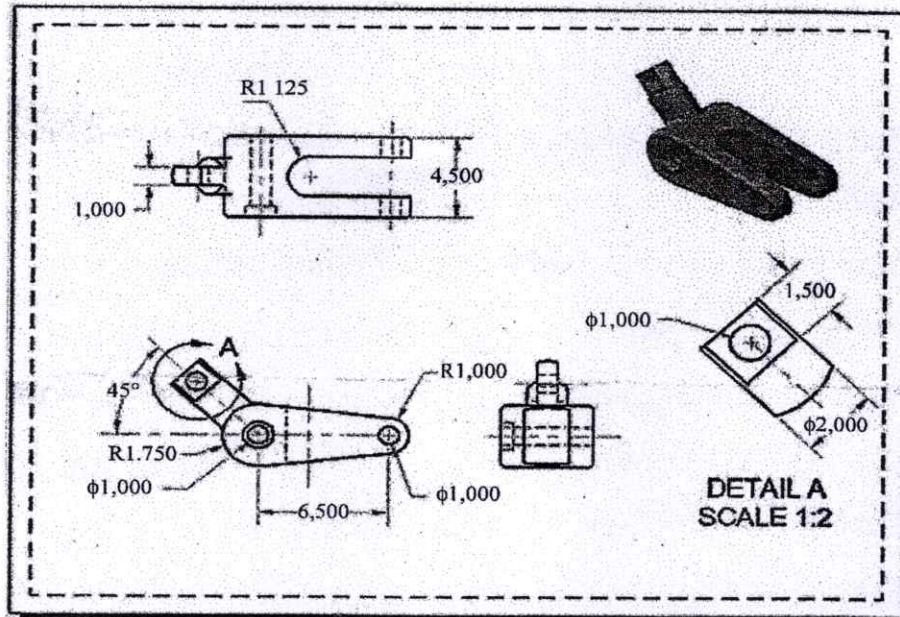
5. (a) Write the steps involve in making 3D print of the given 3D model. Support your answer with flowchart. (CO5)



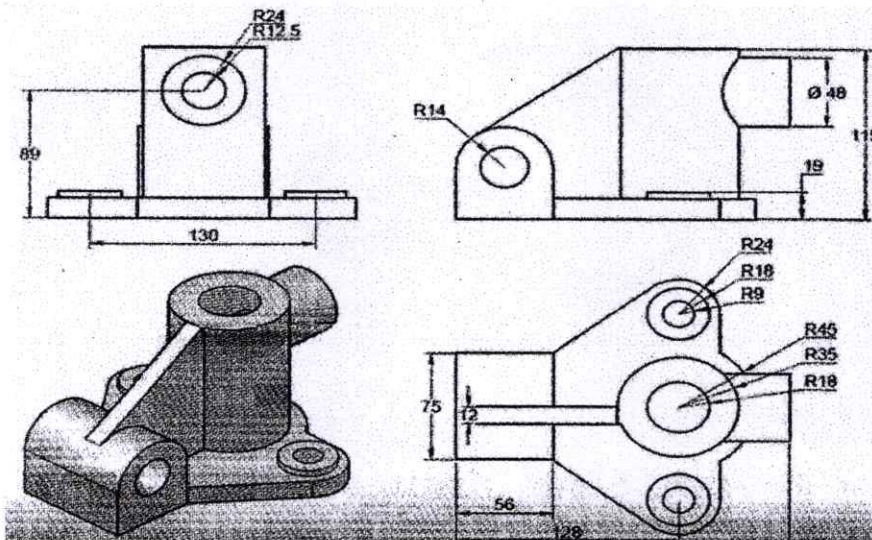
(4)

TBT-505-B

- (b) Write the steps involve in making 3D print of the given 3D model.
Support your answer with flowchart. (CO5)



- (c) Write the steps involved in making 3D print of the given 3D model.
Support your answer with flowchart. (CO5)



TBT-505-B

80

Roll No.

TME-501

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

HEAT AND MASS TRANSFER

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the different modes of heat transfer ? Explain in detail. (CO1)
- (b) What do you mean by thermal contact resistance ? Explain critical thickness of insulation. (CO1)
- (c) Discuss electrical analogy of combined heat conduction and convection in a composite plane wall. (CO1)
2. (a) What are the reasons for using fins on the surface ? Define effectiveness and efficiency of fin. (CO2)
- (b) Establish and expression for temperature distribution in a straight fin of rectangular profile when the fin tip is insulated. (CO2)

P. T. O.

- (c) A longitudinal copper fin ($K=380 \text{ W/m-K}$) 600 mm long and 5 mm diameter is exposed to air stream at 20°C . The convective heat transfer coefficient is $20 \text{ W/m}^2\text{-K}$. If the fin base temperature is 150°C , determine the rate of heat transfer and efficiency of the fin. (CO2)
3. (a) Explain the physical significance of Nusselt number, Grashoff number and Sherwood number. (CO3)
- (b) Explain displacement thickness, momentum thickness and energy thickness used in boundary layer. (CO3)
- (c) Differentiate forced convection and natural convection. Find the Nusselt number of the flowing fluid inside a pipe with heat transfer coefficient (h) = $22 \text{ W/m}^2\text{-k}$, diameter (d) = 0.5 m and conductivity (K) = 25 W/m-k . (CO3)
4. (a) Explain crossed string method for evaluating view factors between infinitely long surfaces. Two infinitely long parallel plates of widths 12 cm and 5 cm are located at a distance 6 cm apart. Determine view factor using crossed string method. (CO4)
- (b) Derive expressions for view factors from any one side to any other side of infinitely long triangular ducts. (CO4)
- (c) Two large parallel planes with emissivities of 0.3 and 0.5 are maintained at temperatures of 527°C and 127°C respectively. A radiation shield having emissivity of 0.05 on both sides is placed between them. Calculate heat transfer rate between them without shield and heat transfer rate between them with shield. (CO4)

(3)

5. (a) Differentiate between parallel flow and counter flow heat exchanger. What would happen to heat transfer if an existing counter flow is replaced by parallel flow heat exchanger ? (CO5)
- (b) What is the heat capacity ? Define the effectiveness of heat exchanger. What would happen to the heat transfer if the heat capacity of fluid increases ? (CO5)
- (c) Hot chemical products ($C_{ph} = 2.5 \text{ kJ/kg K}$) at 600°C and at a flow rate of 30 kg/s are used to heat cold chemical products ($C_{pc} = 4.2 \text{ kJ/kg K}$) at 200°C and at a flow rate of 20 kg/s in a parallel flow heat exchanger. The total heat transfer area is 50 m^2 and the overall heat transfer coefficient may be taken as $1500 \text{ W/m}^2\text{-K}$. Calculate the outlet temperatures of the hot and cold chemical products. (CO5)

Roll No.

TBT-504(B)

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

STEM CELL TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the different methods for fate mapping of stem cells. (CO1)
- (b) Define stem cells and state the different types of stem cells based on their characteristics. (CO1)
- (c) Describe the milestones of advances in stem cell biology. (CO1)
2. (a) Discuss the different types of adult stem cells, transdifferentiation and its applications. (CO2)
- (b) Explain induced pluripotent stem cells. (CO2)
- (c) Briefly describe the potency assays for measuring engraftment potential of hematopoietic and progenitor cells. (CO2)

P. T. O.

(2)

3. (a) Elaborate on the different modes of cell signalling along with the various kinds of receptors. (CO3)
(b) Define cell commitment and differentiation. Describe the factors affecting cell differentiation. (CO3)
(c) Discuss the various stages of embryogenesis. (CO3)
4. (a) Describe embryonic stem cells with emphasis on its production and maintenance. (CO4)
(b) What are stem cell markers ? Discuss the different types of stem cell markers. (CO4)
(c) Explain the process of therapeutic cloning with stem cells. (CO4)
5. (a) Explain tissue engineering and its role in regenerative medicine. (CO5)
(b) Discuss the role of stem cells in the treatment of diabetes. (CO5)
(c) Describe hematopoietic stem cell transplantation (HSCT). (CO5)

Roll No.

TBT-504(A)

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

MARINE BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is Marine Biotechnology ? Discuss some of its applications. (CO1)
(b) Elaborate on some of the economically important finfishes in India. (CO1)
(c) Explain in detail some of the economically important crabs and shrimps in India. (CO1)
2. (a) Elaborate on the different types of extremophiles. (CO2)
(b) What are non-culturable microbes ? Elaborate on the role of microbes in a marine ecosystem. (CO2)
(c) What is eDNA ? Elaborate on the technique of metagenomics. (CO2)

P. T. O.

(2)

3. (a) Explain in detail some of the important bioactive molecules extracted from marine sources. (CO3)
- (b) What are marine toxins ? Elaborate on Conotoxins. (CO3)
- (c) Write in detail about the following marine products : (CO3)
- (i) Tetrodotoxin
- (ii) Antioxidants
4. (a) What is Bioremediation ? Explain how it can be used to restore marine ecosystems. (CO4)
- (b) Elaborate on the methods of remediation of oil spills in marine ecosystems. (CO4)
- (c) What is algal bloom ? Explain what are the disadvantages of algal bloom for the marine ecosystem and how can it be prevented. (CO4)
5. (a) Explain the role of biosensors in marine environmental giving appropriate examples. (CO5)
- (b) Elaborate on some of the diseases associated with cultured shrimps. (CO5)
- (c) Discuss how molecular tools be used to diagnose diseases of marine organisms. (CO5)

Roll No.

TBT-506-B

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

NETWORK BIOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you understand by computer networks ? Discuss various types of network. (CO1)
- (b) Discuss any five terminologies used in network : (CO1)
 - (i) Nodes
 - (ii) Edges
 - (iii) Degree
 - (iv) Shortest path
 - (v) Closeness

P. T. O.

- (c) Elaborate any two real world examples where network biology has been used successfully for solving biological problems. (CO1)
2. (a) Elaborate any *two* protein networks in humans responsible for maintaining life. (CO2)
- (b) Discuss any *one* (1) PM map of *Helicobacter pylori* (2) PPI map of Herpes virus. (CO2)
- (c) Discuss lethality and centrality as major feature of biological networks. (CO2)
3. (a) What are metabolic fluxes ? What is the characteristic of metabolic fluxes in bacterium *Escherichia coli* ? (CO3)
- (b) Discuss how computational analyses and connectivity structure been helpful in drug design. (CO3)
- (c) What do you understand by small world effect for metabolic networks ? What is the giant strong component of metabolic networks ? (CO3)
4. (a) How can gene expression and interaction be represented using network graphs ? Also discuss gene co-expression networks. (CO4)
- (b) Discuss gene-co-expression network for discovery of conserved genetic modules. (CO4)

(3)

(c) What is regulatory network ? Also discuss transcriptional regulatory networks of any one (1) model organism. (CO4)

5. (a) Discuss developmental network construction for understanding evolution of animal body plans. (CO5)

(b) What do you understand by ecological networks ? Discuss food web as example of ecological network. (CO5)

(c) Discuss web of human sexual contacts. What are the applications of this web constructions ? (CO5)

TBT-504A